A Unified Taxonomy of Retrieval- and Knowledge-Augmented AI Systems

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Keywords: Retrieval-Augmented AI, Knowledge-Augmented AI, Knowledge Representation, Neuro-Symbolic AI, Explainable AI, AI Taxonomy

Abstract: Recent advances in Artificial Intelligence have produced a diverse ecosystem of systems that augment large language models with external retrieval, structured knowledge, symbolic reasoning, and tool-mediated computation. However, this landscape remains conceptually fragmented, with frequent ambiguity of architectural designs, reasoning mechanisms, explanation strategies, and epistemic properties. This paper presents a unified multidimensional taxonomy of Retrieval- and Knowledge-Augmented AI Systems, derived from a structured literature analysis of representative contemporary AI systems and architectures. The proposed taxonomy distinguishes orthogonal dimensions including information access mechanisms, knowledge substrates, grounding strategies, reasoning paradigms, explanation strategies, and epistemic properties. Using this framework, we organize modern AI systems into major architectural families and provide a comparative basis for analyzing design trade-offs, explainability, and trustworthiness. The taxonomy aims to establish a coherent conceptual foundation for the systematic analysis and future design of reliable, interpretable, and semantically grounded AI systems.

1 Introduction

The contemporary AI landscape is increasingly characterized by architectures that extend large language models through external retrieval, structured knowledge representations, symbolic reasoning mechanisms, and tool-mediated computation. Architectural paradigms such as Retrieval-Augmented Generation (RAG), GraphRAG, knowledge graph–augmented systems, neuro-symbolic AI, and tool-augmented language models illustrate a broader shift away from purely parametric intelligence toward AI systems whose behavior depends on external evidence, explicit knowledge substrates, and structured reasoning processes. This evolution has significantly expanded the practical capabilities of AI systems, improving factual grounding, access to dynamic information, reasoning transparency, and explainability.

Despite this progress, the research landscape remains conceptually fragmented. Closely related terms such as retrieval-augmented systems, knowledge-augmented architectures, grounded AI, neuro-symbolic AI, explainable AI, and hybrid reasoning systems are often used inconsistently, with blurred distinctions between architectural structures, reasoning mechanisms, explanation strategies, and epistemic properties. Retrieval pipelines are sometimes described as reasoning paradigms, knowledge graphs are occasionally treated as explanation mechanisms, and prompting-based reasoning is frequently conflated with grounded inference. This ambiguity makes systematic comparison difficult and hinders principled analysis of architectural trade-offs, reliability, and trustworthiness.

A central source of this fragmentation is the absence of a coherent conceptual separation between distinct dimensions of AI system design. The mechanisms through which systems access external information are not equivalent to the knowledge substrates over which reasoning operates. Architectural components should be distinguished from inference paradigms, and explanation interfaces should not be confused with epistemic guarantees such as traceability, faithfulness, or verifiability. Likewise, access to retrieved information does not necessarily imply access to structured knowledge, just as reasoning-like outputs do not imply formally grounded inference. Without a unified conceptual framework, comparisons across architectures risk becoming superficial or misleading.

To address this problem, this paper proposes a unified multidimensional taxonomy of Retrieval- and Knowledge-Augmented AI Systems, derived from a structured literature analysis of representative contemporary AI systems and architectures. The proposed taxonomy distinguishes or-

thogonal analytical dimensions including information access mechanisms, knowledge substrates, representation paradigms, grounding strategies, reasoning paradigms, explanation strategies, and epistemic properties. This perspective enables systematic comparison across heterogeneous architectures while preserving conceptual clarity regarding architecture, inference, and trustworthiness.

This paper makes four main contributions. First, we propose a unified multidimensional taxonomy for retrieval- and knowledge-augmented AI systems that integrates architectural, representational, inferential, and epistemic perspectives into a coherent analytical framework. Second, we establish a conceptual separation between architectures, access mechanisms, reasoning paradigms, explanation strategies, and epistemic properties, clarifying distinctions that are frequently blurred in the literature. Third, we provide a comparative framework for positioning contemporary AI architectures according to these dimensions, enabling systematic cross-architectural analysis. Fourth, we discuss implications for the design of future trustworthy, interpretable, and semantically grounded AI systems.

The remainder of this paper is organized as follows. Section 2 presents the conceptual foundations underlying the proposed taxonomy. Section 3 introduces the taxonomy and the structured literature analysis methodology. Section 4 provides a comparative analysis of representative architectures using the proposed framework, and Section 5 concludes the paper and points directions for future work.

2 Foundations

A coherent taxonomy of Retrieval- and Knowledge-Augmented AI Systems requires clear conceptual distinctions between information access, knowledge representation, reasoning processes, and explainability. In contemporary AI literature, these notions are often used with overlapping meanings, which makes systematic comparison across architectures difficult. This section establishes the conceptual foundations required for the proposed taxonomy.

2.1 Information, Knowledge, and External Evidence

A fundamental distinction concerns the difference between information and knowledge. Information refers to representational content accessible to a system, including textual documents, vectorized artifacts, database records, or other externally available resources. Retrieval-augmented architectures typically operate over such information spaces by selecting relevant content based on lexical or semantic similarity (Lewis et al., 2020).

Knowledge, in contrast, implies a higher degree of semantic organization that supports interpretation and inference. Knowledge graphs provide a representative example of explicit knowledge structures, since they encode entities and semantic relations in a machine-interpretable form (Hogan et al., 2021). This distinction is important because access to information does not necessarily imply access to structured knowledge suitable for reasoning.

Modern language models introduce an additional consideration. Large neural models may encode factual associations within their parameters as a result of large-scale training, but such representations remain implicit and difficult to inspect directly (Brown et al., 2020). As a result, architectures that rely on parametric associations differ fundamentally from systems that incorporate explicitly structured external knowledge.

2.2 Information Access and Knowledge Substrates

AI architectures also differ in how they access external resources. Dense retrieval mechanisms identify relevant documents by mapping queries and passages into shared embedding spaces and selecting semantically similar candidates (Karpukhin et al., 2020). In structured environments, information access may instead occur through formal querying mechanisms, such as text-to-SQL or neural semantic parsing approaches that translate natural language into executable database queries (Yin and Neubig, 2017). Such architectures provide direct access to structured organizational knowledge through executable query generation rather than similarity-based retrieval.

A different access paradigm involves external computational tools. Tool-augmented architectures allow language models to invoke external functions, APIs, or computational services during inference rather than relying exclusively on internal model knowledge (Schick et al., 2023). These access mechanisms differ substantially in their assumptions regarding relevance, structure, and computational interaction.

Beyond access mechanisms, architectures also differ in their underlying knowledge substrates. Some systems operate over unstructured corpora or vector stores, while others rely on explicitly structured knowledge repositories such as relational databases, knowledge graphs, ontologies, or symbolic reason-

ing substrates. In enterprise environments, natural-language interfaces over structured databases represent an important form of knowledge augmentation, since user intent is grounded in formal schema structures and translated into executable access over explicit knowledge repositories (Pinheiro et al., 2023). Neuro-symbolic systems exemplify architectures that combine learned representations with structured reasoning components in order to integrate statistical learning with formal inference capabilities (Hitzler et al., 2022). Distinguishing access mechanisms from knowledge substrates is therefore essential for meaningful architectural comparison.

2.3 Grounding, Reasoning, and Explainability

Grounding concerns the relationship between internal representations and external referents or evidence. The symbol grounding problem established the foundational observation that symbolic manipulation alone does not guarantee semantic meaning unless representations are connected to entities or states of affairs beyond the formal system itself (Harnad, 1990). In modern AI architectures, grounding determines whether outputs can be linked to external evidence rather than merely generated from internal statistical associations.

Reasoning refers to the inferential processes through which systems derive conclusions from available representations. Recent work on chain-of-thought prompting demonstrates that large language models can generate multi-step reasoning-like outputs under suitable prompting conditions (Wei et al., 2022). However, such outputs should not automatically be interpreted as equivalent to formal inference, since their generation may reflect statistical pattern completion rather than logically constrained reasoning.

Explainability introduces a distinct dimension. Interpretability research has shown that explanations generated by AI systems may appear plausible without faithfully reflecting the mechanisms that produced the output (Lipton, 2018). Similarly, faithful explanation requires that explanatory accounts correspond to the actual computational basis of the decision rather than post-hoc rationalizations (Jacovi and Goldberg, 2020). These distinctions are central to the proposed taxonomy, since architectural transparency, inferential reasoning, evidential grounding, and explanation should be treated as analytically separate dimensions rather than interchangeable properties.

3 Taxonomy of Retrieval- and Knowledge-Augmented AI Systems

This section presents the proposed taxonomy of Retrieval- and Knowledge-Augmented AI Systems, derived from the conceptual distinctions established in the previous section. The objective is not merely to enumerate existing architectures, but to provide a structured analytical framework that enables systematic comparison across heterogeneous approaches. Contemporary AI systems increasingly combine external retrieval, structured knowledge representations, explicit reasoning mechanisms, and computational tools, yet these capabilities are often discussed using overlapping terminology that obscures meaningful architectural distinctions.

The proposed taxonomy organizes this landscape along orthogonal analytical dimensions that separate how systems access information, how knowledge is represented, how reasoning is performed, and how epistemic guarantees are established. Based on this framework, representative architectures are grouped into major families that reflect recurring structural design patterns rather than strict mutually exclusive categories.

3.1 Structured Literature Analysis Methodology

The taxonomy was derived from a structured literature analysis of representative contemporary AI systems published between 2018 and 2025, a period characterized by the rapid emergence of retrieval-augmented language models, graph-based retrieval architectures, knowledge-grounded systems, neuro-symbolic AI, and tool-augmented language agents. The analysis considered influential peer-reviewed works from major artificial intelligence, natural language processing, and data management venues, complemented by seminal preprints that introduced paradigms later broadly adopted by the research community.

Selection criteria prioritized architectures that explicitly incorporate at least one of the following properties: dynamic retrieval from external information sources, explicit structured knowledge representations, formal reasoning mechanisms, or interaction with external computational tools during inference. Purely parametric language models without architectural relevance to retrieval, reasoning, or explainability were excluded unless they contributed foundational mechanisms later adopted by augmented archi-

tectures.

Rather than aiming for exhaustive coverage, the analysis emphasizes structurally representative systems that illustrate major architectural patterns and conceptual design choices. This approach prioritizes analytical clarity and taxonomic consistency over exhaustive enumeration.

3.2 Taxonomic Dimensions

The proposed taxonomy organizes AI systems along dimensions that capture structurally meaningful differences in design and epistemic behavior. The first dimension concerns *information access mechanisms*, including dense retrieval, structured querying, graph traversal, memory access, and external tool invocation. These mechanisms determine how systems obtain information beyond their parametric representations. The second dimension concerns *knowledge substrates*, referring to the representational structures over which systems operate. These range from unstructured textual corpora and vector stores to explicitly structured repositories such as relational databases, knowledge graphs, ontologies, symbolic rule bases, or hybrid semantic reasoning substrates.

The third dimension concerns *grounding strategy*, capturing how internal representations are aligned with external evidence or semantic referents. Grounding may occur prior to inference, during iterative reasoning, or as a validation mechanism applied after generation. The fourth dimension concerns *reasoning paradigm*, including neural reasoning, symbolic deduction, probabilistic inference, constraint-based reasoning, planning-based reasoning, or hybrid combinations. The fifth dimension concerns *explainability strategy*, distinguishing systems that provide no explicit explanations, post-hoc justifications, or explanations intrinsically linked to evidence or inference processes. Finally, the taxonomy considers *epistemic properties*, including traceability, faithfulness, verifiability, and evidential transparency. These properties are treated as evaluative characteristics rather than architectural components. Based on these dimensions, the taxonomy organizes representative systems into five major architectural families: retrieval-augmented architectures, graph-based retrieval architectures, knowledge-augmented architectures, symbolic and neuro-symbolic knowledge architectures, and tool-augmented architectures.

Figure 1 summarizes the proposed taxonomy by organizing contemporary AI architectures into three macro-level conceptual groupings. Retrieval-oriented architectures emphasize dynamic access to external information, including classical retrieval-augmented systems and graph-based retrieval approaches that exploit structural connectivity during evidence discovery. Knowledge-oriented architectures rely on explicitly structured knowledge substrates, including relational databases, knowledge graphs, ontologies, and symbolic reasoning repositories, ranging from knowledge-augmented systems grounded in formal representations to symbolic and neuro-symbolic architectures capable of explicit inference. Tool-augmented systems are treated separately as capability-extension architectures, since their defining characteristic is dynamic interaction with external computational resources rather than information retrieval or structured knowledge representation. Across all architectural families, the taxonomy applies a common analytical lens based on access mechanisms, knowledge substrates, grounding strategies, reasoning paradigms, explainability strategies, and epistemic properties.

3.3 Retrieval-Augmented Architectures

Retrieval-augmented architectures constitute a major class of AI systems in which model outputs are conditioned on information dynamically retrieved from external sources. Their defining architectural principle is the explicit separation between parametric language generation and external information access, allowing models to consult non-parametric memory during inference rather than relying exclusively on internally encoded associations. The modern formulation of this paradigm was established by Retrieval-Augmented Generation (RAG), which introduced a modular architecture combining dense retrieval with neural sequence generation (Lewis et al., 2020). This design established retrieval as an explicit architectural component rather than a preprocessing convenience, enabling systems to dynamically incorporate external evidence into response generation.

Subsequent architectures explored variations of this principle. Fusion-in-Decoder extended the retrieve-then-generate paradigm by integrating retrieved passages directly into the decoder process, improving evidence aggregation during generation (Izacard and Grave, 2021). RETRO further strengthened the role of external memory by treating retrieved context as an extension of model computation itself (Borgeaud et al., 2022). Atlas demonstrated how retrieval augmentation can support few-shot adaptation without requiring parameter updates (Izacard et al., 2023).

Other systems focused on preserving architectural simplicity while improving retrieval effectiveness. (Shi et al., 2024) showed that retrieval augmen-

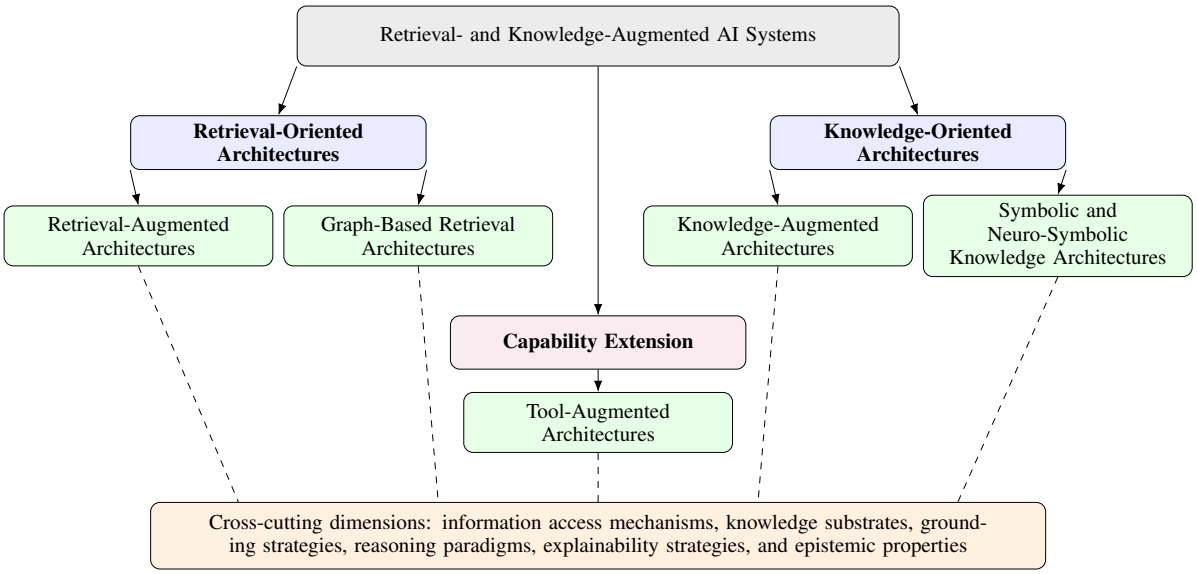


Figure 1: Taxonomy of retrieval- and knowledge-augmented AI systems. The taxonomy distinguishes retrieval-oriented architectures, knowledge-oriented architectures, and capability-extension architectures, while analyzing all classes through cross-cutting dimensions related to access, representation, grounding, reasoning, explainability, and epistemic properties.

tation can be applied externally without modifying the base language model architecture. Self-RAG introduced adaptive retrieval policies in which the system determines when retrieval is necessary and uses retrieved evidence to improve response quality (Asai et al., 2024). kNN-LM explored fine-grained retrieval by interpolating predictions with nearest-neighbor information stored in external datastores (Khandelwal et al., 2019).

Despite architectural diversity, these systems share a common structural principle: retrieval serves as an external information access mechanism that complements parametric generation. This design improves updateability, scalability, and evidential traceability. However, retrieval alone does not guarantee grounded reasoning, semantic consistency, or faithful explanation, since these properties depend on how retrieved information is selected, integrated, and validated.

3.4 Graph-Based Retrieval Architectures

Graph-based retrieval architectures extend traditional retrieval paradigms by organizing retrieved information within structured relational spaces rather than treating documents as independent units. In these systems, retrieval becomes a process of graph exploration in which relevant evidence is identified through structural connectivity among entities, concepts, passages, or contextual nodes. Unlike conventional retrieval pipelines based solely on similarity ranking, graph-based retrieval systems exploit explicit relationships among information units to support contextual expansion and multi-hop evidence discovery. This allows systems to assemble distributed evidence that would be difficult to recover through isolated document retrieval alone.

This architectural paradigm gained visibility through GraphRAG, which organizes retrieved information into graph structures that support structured exploration and contextual aggregation during generation (Edge et al., 2024). By introducing graph connectivity into retrieval, these systems improve contextual coherence and enable reasoning over relational evidence rather than isolated passages. Related approaches integrate graph neural networks to propagate information across structured retrieval spaces prior to generation (Wu et al., 2020). Other architectures combine dense retrieval with graph expansion, using initial retrieval steps to identify seed nodes that are subsequently expanded through semantic neighbors or relational traversal.

Graph-based retrieval architectures occupy an intermediate position between classical retrieval systems and explicit knowledge-based systems. Like retrieval architectures, they preserve dynamic access to external information. However, by introducing structural organization, they partially approximate knowledge-oriented reasoning capabilities, particularly in tasks requiring relational context, evidence chaining, or multi-hop inference.

Nevertheless, graph-based retrieval should be dis-

tinguished from explicit knowledge augmentation. While graph organization improves retrieval structure, these systems do not necessarily operate over formally curated semantic knowledge representations. Their graph structures may emerge from document connectivity or extracted relations rather than from explicit symbolic knowledge engineering.

3.5 Knowledge-Augmented Architectures

Knowledge-augmented architectures constitute a class of AI systems in which inference is explicitly supported by structured knowledge representations rather than relying solely on retrieved textual evidence. Their defining characteristic is the incorporation of semantically organized external knowledge substrates that provide machine-interpretable representations of entities, relations, constraints, or domain concepts. A representative form of this paradigm is the use of knowledge graphs as persistent semantic memory. Knowledge graphs organize information through explicit relational structures that support semantic interoperability, compositional querying, and structured reasoning over entities and relations (Hogan et al., 2021). Because these structures encode explicit semantics rather than statistical associations alone, they provide stronger foundations for explainability, constraint enforcement, and evidence traceability.

Knowledge-augmented architectures differ fundamentally from retrieval-augmented systems. Retrieval mechanisms primarily provide access to relevant information artifacts, whereas knowledge-augmented systems operate over explicitly structured knowledge that can support semantic interpretation and formal inference. This distinction becomes particularly important in applications requiring consistency checking, compositional reasoning, or explicit domain semantics. Several architectural patterns have emerged within this family. Query-driven architectures, including text-to-SQL and semantic querying systems, translate natural-language inputs into executable structured queries evaluated against explicit knowledge repositories such as relational databases and semantic knowledge bases (Seabra et al., 2025). In these systems, user intent is grounded in formal schema structures, enabling direct interaction with structured organizational knowledge through executable query generation. Other systems inject graph-derived or schema-grounded knowledge into generation contexts so that explicit semantic relations constrain downstream responses. Hybrid approaches increasingly combine dense retrieval with structured knowledge access, allowing retrieval mechanisms to identify candidate knowledge regions while semantic reasoning operates over explicitly structured representations.

The principal advantage of knowledge augmentation lies in epistemic structure. Because evidence is represented explicitly, outputs can often be linked to identifiable entities, relations, or reasoning paths, enabling stronger forms of traceability and verification than architectures based solely on similarity-driven retrieval. However, these benefits come at the cost of knowledge engineering complexity, including schema construction, semantic alignment, ontology maintenance, and knowledge base evolution. From a taxonomic perspective, knowledge-augmented architectures represent a distinct design family because they treat structured knowledge as a first-class inferential substrate rather than merely as contextual information injected into generation pipelines.

3.6 Symbolic and Neuro-Symbolic Knowledge Architectures

Symbolic and neuro-symbolic knowledge architectures extend knowledge augmentation by incorporating explicit reasoning mechanisms over structured representations. While knowledge-augmented systems provide semantic substrates for inference, symbolic and neuro-symbolic architectures introduce formal inferential processes capable of operating over those substrates.

Symbolic architectures rely on explicit knowledge representations such as logical rules, ontologies, description logic structures, or formal knowledge bases, enabling inference procedures with well-defined semantics and stronger guarantees regarding consistency and interpretability. These systems are particularly suitable for applications requiring rule enforcement, logical verification, or transparent reasoning over explicitly modeled knowledge.

Neuro-symbolic architectures attempt to combine the complementary strengths of neural and symbolic paradigms. Neural models provide scalability, pattern recognition, and language understanding, while symbolic components contribute explicit semantics, compositional reasoning, and formal inferential structure (Hitzler et al., 2022). This integration seeks to overcome limitations of purely neural systems in tasks requiring structured reasoning or logical consistency.

Different integration strategies have been proposed. One common approach uses neural components for language understanding, semantic parsing, or representation extraction, followed by symbolic inference over structured knowledge bases. Another ap-

Table 1: Comparative classification of retrieval-oriented representative architectures.

System	Architectural Family	Key Characteristics
RAG	Retrieval-Augmented	Access: Dense retrieval; Knowledge substrate: Text corpus; Grounding: Pre-inference contextual grounding; Reasoning: Neural generation; Explainability: Source attribution.
FiD	Retrieval-Augmented	Access: Dense retrieval; Knowledge substrate: Text corpus; Grounding: Pre-inference evidence aggregation; Reasoning: Neural generation; Explainability: Limited evidence traceability.
RETRO	Retrieval-Augmented	Access: Dense retrieval; Knowledge substrate: External text memory; Grounding: Retrieval-conditioned generation; Reasoning: Neural generation; Explainability: Limited source traceability.
Atlas	Retrieval-Augmented	Access: Dense retrieval; Knowledge substrate: Text corpus; Grounding: Retrieval-supported adaptation; Reasoning: Neural generation; Explainability: Moderate evidence traceability.
RePlug	Retrieval-Augmented	Access: Dense retrieval; Knowledge substrate: External text corpus; Grounding: External contextual grounding; Reasoning: Neural generation; Explainability: Moderate source attribution.
Self-RAG	Retrieval-Augmented	Access: Adaptive retrieval; Knowledge substrate: Text corpus; Grounding: Iterative self-grounding; Reasoning: Neural reflective reasoning; Explainability: Improved evidence awareness.
kNN-LM	Retrieval-Augmented	Access: Nearest-neighbor retrieval; Knowledge substrate: Vector datastore; Grounding: Memory-conditioned generation; Reasoning: Neural generation; Explainability: Limited traceability.
GraphRAG	Graph-Based Retrieval	Access: Graph traversal + retrieval; Knowledge substrate: Relational document graph; Grounding: Structured contextual grounding; Reasoning: Multi-hop contextual reasoning; Explainability: Path-based traceability.
HippoRAG	Graph-Based Retrieval	Access: Associative graph retrieval; Knowledge substrate: Graph memory; Grounding: Memory graph grounding; Reasoning: Multi-hop associative reasoning; Explainability: Moderate path traceability.

proach incorporates symbolic constraints directly into neural learning or decoding processes, enabling models to produce outputs consistent with predefined logical or semantic rules (Dong et al., 2019).

A broader class of hybrid architectures integrates external reasoning engines into language-model workflows. In such systems, symbolic rule engines, logic programming systems, or ontology reasoners perform explicit reasoning over structured knowledge, while neural models handle ambiguity, contextual interpretation, or language interaction. This separation allows architectures to combine statistical flexibility with formal inferential transparency.

These architectures provide the strongest support for explainability and epistemic verification among the classes considered in this taxonomy. Because reasoning operates over explicit structures, inferential steps can often be inspected, validated, or formally analyzed. However, integration remains challenging, since reliable mapping between natural language inputs and symbolic representations requires robust semantic alignment mechanisms.

From the perspective of this taxonomy, symbolic and neuro-symbolic architectures represent the strongest realization of knowledge augmentation, since they combine explicit knowledge representation with formal reasoning processes rather than using structured knowledge solely as contextual support.

3.7 Tool-Augmented Architectures

Tool-augmented architectures represent a distinct class of AI systems in which language models extend their capabilities through interaction with external computational resources during inference. Rather than relying exclusively on internal parametric knowledge or passive information retrieval, these systems dynamically invoke external tools such as search engines, databases, APIs, calculators, or domain-specific computational services.

In these architectures, the language model functions as an orchestration layer that determines when external computation or information access is required. Toolformer demonstrated that language models can learn to identify opportunities for tool usage and integrate tool outputs into downstream reasoning without major architectural redesign (Schick et al., 2023). This established tool invocation as an explicit mechanism for extending model competence beyond internal representations.

ReAct further expanded this paradigm by interleaving reasoning steps with external actions, enabling models to iteratively gather information and update reasoning trajectories through interaction with external environments (Yao et al., 2022). Such architectures blur the boundary between reasoning and action, transforming language models from passive generators into active computational agents. Tool augmentation differs fundamentally from knowledge augmentation. Knowledge-augmented systems rely on persistent semantic representations that explicitly encode structured knowledge. Tool-augmented systems, by contrast, extend capability through dynamic interaction with external computational processes, which may or may not involve structured knowledge representations.

This flexibility makes tool-augmented architectures particularly attractive for tasks requiring com-

Table 2: Comparative classification of knowledge-oriented and tool-augmented representative architectures.

System	Architectural Family	Key Characteristics
KGQA	Knowledge-Augmented	Access: Structured querying; Knowledge substrate: Knowledge graph; Grounding: Semantic grounding; Reasoning: Symbolic query execution; Explainability: High traceability.
KnowBERT	Knowledge-Augmented	Access: Entity-aware retrieval; Knowledge substrate: Knowledge graph; Grounding: Entity grounding; Reasoning: Neural knowledge integration; Explainability: Moderate semantic traceability.
K-BERT	Knowledge-Augmented	Access: Knowledge injection; Knowledge substrate: Knowledge graph; Grounding: Structured semantic grounding; Reasoning: Knowledge-constrained neural reasoning; Explainability: Moderate interpretability.
Semantic RAG	Knowledge-Augmented / Hybrid	Access: Retrieval + semantic retrieval; Knowledge substrate: Structured semantic knowledge; Grounding: Semantic evidence grounding; Reasoning: Hybrid contextual reasoning; Explainability: Improved semantic traceability.
Text-to-SQL	Knowledge-Augmented	Access: Structured querying; Knowledge substrate: Relational database schema; Grounding: Schema and execution grounding; Reasoning: Neural semantic parsing or constrained query generation; Explainability: High executable traceability.
DeepProbLog	Symbolic / Neuro-Symbolic	Access: Hybrid access; Knowledge substrate: Probabilistic symbolic knowledge; Grounding: Explicit semantic grounding; Reasoning: Hybrid formal reasoning; Explainability: High interpretability.
Logic Tensor Networks	Symbolic / Neuro-Symbolic	Access: Hybrid access; Knowledge substrate: Logical semantic constraints; Grounding: Formal semantic grounding; Reasoning: Neural-symbolic reasoning; Explainability: High interpretability.
Neural Theorem Provers	Symbolic / Neuro-Symbolic	Access: Hybrid access; Knowledge substrate: Logical rule base; Grounding: Formal symbolic grounding; Reasoning: Differentiable theorem proving; Explainability: High inferential transparency.
Toolformer	Tool-Augmented	Access: Tool invocation; Knowledge substrate: External computational tools; Grounding: Dynamic execution grounding; Reasoning: Tool-assisted reasoning; Explainability: Limited process transparency.
ReAct	Tool-Augmented	Access: Tool interaction + retrieval; Knowledge substrate: External environments; Grounding: Iterative action grounding; Reasoning: Reasoning + action loops; Explainability: Process traceability.
MRKL	Tool-Augmented	Access: Modular tool routing; Knowledge substrate: External expert modules; Grounding: Tool-mediated execution grounding; Reasoning: Modular reasoning orchestration; Explainability: Moderate process transparency.

putation, real-time information access, or interaction with heterogeneous systems. However, these benefits introduce new design challenges, including tool selection, interface reliability, error propagation, and coordination between language reasoning and external execution. Within the proposed taxonomy, tool-augmented architectures are treated as a distinct family because their defining characteristic is computational extensibility rather than retrieval structure or explicit knowledge representation.

4 Comparative Analysis and Taxonomy Validation

To assess the analytical utility of the proposed taxonomy, this section applies the framework to a representative set of contemporary AI architectures spanning the major categories identified in Section 3. The objective is not to provide exhaustive coverage of the literature, but to demonstrate how the proposed dimensions support systematic classification and cross-architectural comparison.

4.1 Comparative Classification of Representative Systems

Tables 1, 2 and 3 apply the proposed taxonomy to representative systems drawn from retrieval-augmented, graph-based retrieval, knowledge-augmented, symbolic/neuro-symbolic, and tool-augmented architectural families.

The comparison highlights that architectures often differ not only in their information access mechanisms, but also in the nature of their knowledge substrates, grounding strategies, reasoning paradigms, and explainability properties. This reinforces the central claim of this work that architectural labels alone are insufficient for meaningful comparison unless these dimensions are explicitly separated.

4.2 Cross-Architectural Analysis

The comparative classification reveals several important structural patterns. First, retrieval-oriented architectures emphasize scalability and access to dynamic information, but their epistemic guarantees remain limited when reasoning is performed primarily through neural generation over retrieved context. While retrieval improves factual grounding, it does not inherently provide semantic consistency or formal inferential transparency.

Second, graph-based retrieval architectures par-

Table 3: Comparative heatmap of representative AI architectures.

System	Retrieval	Structured Knowledge	Symbolic Reasoning	Tool Use	Grounding	Explainability
RAG	✓	–	–	–	∼	∼
FiD	✓	–	–	–	∼	∼
RETRO	✓	–	–	–	∼	–
Atlas	✓	–	–	–	∼	∼
RePlug	✓	–	–	–	∼	∼
Self-RAG	✓	–	–	–	✓	∼
kNN-LM	✓	–	–	–	∼	–
GraphRAG	✓	∼	–	–	✓	✓
HippoRAG	✓	∼	–	–	✓	∼
KGQA	–	✓	✓	–	✓	✓
KnowBERT	∼	✓	–	–	✓	∼
K-BERT	∼	✓	–	–	✓	∼
Semantic RAG	✓	✓	–	–	✓	✓
Text-to-SQL	–	✓	∼	–	✓	✓
DeepProbLog	–	✓	✓	–	✓	✓
LTN	–	✓	✓	–	✓	✓
NTP	–	✓	✓	–	✓	✓
Toolformer	–	–	–	✓	∼	–
ReAct	∼	–	–	✓	✓	∼
MRKL	–	–	–	✓	✓	∼

✓ = strong presence; ∼ = partial presence; – = absent.

tially bridge retrieval and knowledge-oriented reasoning by introducing relational structure into evidence access. These systems improve contextual coherence and multi-hop evidence discovery, but their graph structures do not necessarily constitute formally curated semantic knowledge. Third, knowledge-augmented and symbolic architectures provide stronger epistemic properties because reasoning operates over explicitly structured semantic substrates. This includes not only knowledge graphs and symbolic rule systems, but also query-driven architectures such as text-to-SQL systems, in which natural-language intent is grounded in formal database schemas and executable query semantics. The presence of formal entities, relations, and inferential mechanisms enables higher levels of traceability, interpretability, and verifiability. However, these benefits often require substantial knowledge engineering effort and reduced flexibility compared with purely neural architectures.

Finally, tool-augmented systems emphasize capability extension rather than explicit knowledge representation. Their strength lies in dynamic interaction with computational environments, but explainability and epistemic guarantees depend heavily on orchestration quality and tool reliability. These observations demonstrate that the proposed taxonomy supports distinctions that are difficult to capture through conventional architectural labels alone. In particular, separating retrieval augmentation from explicit knowledge augmentation clarifies important differences in reasoning behavior, grounding strategy, and explainability across modern AI systems.

5 Conclusions and Future Work

This paper presented a unified multidimensional taxonomy of Retrieval- and Knowledge-Augmented AI Systems, providing a structured analytical framework for organizing a research landscape that remains conceptually fragmented despite its rapid evolution. By explicitly separating information access mechanisms, knowledge substrates, grounding strategies, reasoning paradigms, explainability strategies, and epistemic properties, the proposed taxonomy enables more precise comparison across heterogeneous AI architectures that are often described using overlapping or inconsistent terminology.

The comparative analysis demonstrated that conventional architectural labels alone are insufficient to capture meaningful differences among contemporary systems. In particular, the distinction between retrieval augmentation and explicit knowledge augmentation proved essential for understanding differences in grounding behavior, inferential transparency, explainability, and epistemic reliability. Retrieval-oriented architectures offer scalability and dynamic access to information, while knowledge-oriented and symbolic architectures provide stronger support for semantic consistency, traceability, and formal reasoning. Tool-augmented systems, in turn, expand computational capabilities through dynamic interaction with external environments.

Several directions emerge for future research. First, the increasing convergence between retrieval, structured knowledge, symbolic reasoning, and tool orchestration suggests the need for more refined taxonomic models capable of representing hybrid ar-

chitectures whose boundaries are becoming progressively less distinct. Second, future work should investigate evaluation frameworks that operationalize epistemic properties such as faithfulness, verifiability, and grounding reliability, enabling more systematic assessment of trustworthy AI architectures. Third, the integration of explicit semantic knowledge representations with neural reasoning remains an important open challenge, particularly in architectures that seek to balance flexibility, scalability, and formal interpretability. Finally, emerging agentic AI systems raise new taxonomic questions regarding dynamic planning, tool coordination, and adaptive multi-step reasoning, which may require extending the taxonomy beyond current architectural paradigms.

By establishing a coherent conceptual foundation for analyzing retrieval- and knowledge-augmented AI systems, this work aims to support the development of future AI architectures that are not only more capable, but also more reliable, interpretable, and epistemically grounded.

REFERENCES

Asai, A., Wu, Z., Wang, Y., Sil, A., and Hajishirzi, H. (2024). Self-rag: Learning to retrieve, generate, and critique through self-reflection. In *International conference on learning representations*, volume 2024, pages 9112–9141.

Borgeaud, S., Mensch, A., Hoffmann, J., Cai, T., Rutherford, E., Millican, K., Van Den Driessche, G. B., Lespiau, J.-B., Damoc, B., Clark, A., et al. (2022). Improving language models by retrieving from trillions of tokens. In *International conference on machine learning*, pages 2206–2240. PMLR.

Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., et al. (2020). Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901.

Dong, H., Mao, J., Lin, T., Wang, C., Li, L., and Zhou, D. (2019). Neural-symbolic learning and reasoning: A survey and interpretation. *arXiv preprint arXiv:1912.06088*.

Edge, D., Trinh, H., Cheng, N., Bradley, J., Chao, A., Mody, A., Truitt, S., Metropolitansky, D., Ness, R. O., and Larson, J. (2024). From local to global: A graph rag approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*.

Harnad, S. (1990). The symbol grounding problem. *Physica D: Nonlinear Phenomena*, 42(1-3):335–346.

Hitzler, P., Sarker, M., Besold, T., Garcez, A., Bader, S., Bowman, H., Domingos, P., Hitzler, P., Kühnberger, K., Lamb, L., et al. (2022). Neural-symbolic learning and reasoning: A survey and interpretation. *Frontiers in artificial intelligence and applications*, 342:1–51.

Hogan, A., Blomqvist, E., Cochez, M., d'Amato, C., Melo, G. D., Gutierrez, C., Kirrane, S., Gayo, J. E. L., Navigli, R., Neumaier, S., et al. (2021). Knowledge graphs. *ACM Computing Surveys (Csur)*, 54(4):1–37.

Izacard, G. and Grave, E. (2021). Leveraging passage retrieval with generative models for open domain question answering. In *Proceedings of the 16th conference of the european chapter of the association for computational linguistics: main volume*, pages 874–880.

Izacard, G., Lewis, P., Lomeli, M., Hosseini, L., Petroni, F., Schick, T., Dwivedi-Yu, J., Joulin, A., Riedel, S., and Grave, E. (2023). Atlas: Few-shot learning with retrieval augmented language models. *Journal of Machine Learning Research*.

Jacovi, A. and Goldberg, Y. (2020). Towards faithfully interpretable nlp systems: How should we define and evaluate faithfulness? In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4198–4205. Association for Computational Linguistics.

Karpukhin, V., Oguz, B., Min, S., Lewis, P., Wu, L., Edunov, S., Chen, D., and Yih, W.-t. (2020). Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 conference on empirical methods in natural language processing (EMNLP)*, pages 6769–6781.

Khandelwal, U., Levy, O., Jurafsky, D., Zettlemoyer, L., and Lewis, M. (2019). Generalization through memorization: Nearest neighbor language models. *arXiv preprint arXiv:1911.00172*.

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.-t., Rocktäschel, T., et al. (2020). Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems*, 33:9459–9474.

Lipton, Z. C. (2018). The mythos of model interpretability: In machine learning, the concept of interpretability is both important and slippery. *Queue*, 16(3):31–57.

Pinheiro, J., Victorio, W., Nascimento, E., Seabra, A., Izquierdo, Y., García, G., Coelho, G. M., Lemos, M., Leme, L. A. P. P., Furtado, A., et al. (2023). On the construction of database interfaces based on large language models. In *WEBIST*, pages 373–380.

Schick, T., Dwivedi-Yu, J., Dessì, R., Raileanu, R., Lomeli, M., Hambro, E., Zettlemoyer, L., Cancedda, N., and Scialom, T. (2023). Toolformer: Language models can teach themselves to use tools. *Advances in neural information processing systems*, 36:68539–68551.

Seabra, A., Cavalcante, C., and Lifschitz, S. (2025). Enhancing explainability in ai-powered data retrieval systems. In *IFIP International Conference on Artificial Intelligence Applications and Innovations*, pages 405–418. Springer.

Shi, W., Min, S., Yasunaga, M., Seo, M., James, R., Lewis, M., Zettlemoyer, L., and Yih, W.-t. (2024). Replug: Retrieval-augmented black-box language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 8371–8384.

Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., Le, Q. V., Zhou, D., et al. (2022). Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837.

Wu, Z., Pan, S., Chen, F., Long, G., Zhang, C., and Yu, P. S. (2020). A comprehensive survey on graph neural networks. *IEEE transactions on neural networks and learning systems*, 32(1):4–24.

Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., and Cao, Y. (2022). React: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*.

Yin, P. and Neubig, G. (2017). A syntactic neural model for general-purpose code generation. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 440–450.